

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

Academic Year: 2025-26	
Semester: III	Course Code:CSAIML302T
Name of the Course: DATABASE SYSTEMS	Tutorials/week: 5Hrs
Teaching Scheme: TH: 03Hours/Week	Credit:04
Examination Scheme:	
Formative Assessment: 50 Marks Summative Assessment: 100 Marks	

Question Bank

CO- Course Outcome, BL- Bloom's Level (L1, L2, L3, L4, L5, L6)

S.No.	Unit	Question	Mark s	CO	B L	Asked in Competitive Exam
UNIT - 1						
1	1	What is the primary purpose of a DBMS? a) Data visualization b) Efficient data storage and retrieval c) Hardware control d) Operating system development	1	CO 1	L1	GATE 2000
2	1	Which level of data abstraction describes how data is stored physically? a) Logical level b) Physical level c) View level d) Conceptual level	1	CO 1	L2	NET 2002
3	1	What does "data independence" refer to in a DBMS? a) Accessing data from multiple locations b) Ability to change schema at one level without affecting other levels c) Storing data without redundancy d) Keeping data secure	1	CO 1	L2	GATE 1992
4	1	Which data model is based on tables with rows and columns? a) Hierarchical model b) Network model c) Relational model d) Object-oriented model	1	CO 1	L1	UGC NET 1998
5	1	In the Entity-Relationship model, what represents a "thing" in the real world? a) Attribute b) Entity	1	CO 1	L1	GATE 2019

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

		c) Relationship d) Schema				
6	1	What is an "instance" in a database? a) A table structure b) A type of attribute c) A snapshot of data at a particular moment d) A type of relationship	1	CO 1	L2	NET 2021
7	1	Which language is used to define database schemas? a) DML b) DDL c) TCL d) SQL	1	CO 1	L1	GATE 2023
8	1	What does a "weak entity" depend on? a) A strong entity and a foreign key b) Itself c) Multiple attributes d) Redundancy	1	CO 1	L2	GATE 2020
9	1	Which of the following is NOT a type of database user? a) Application programmer b) End user c) Operating system d) Database administrator	1	CO 1	L3	NET 2020
10	1	What is the role of an "attribute" in an ER diagram? a) Connects entities b) Represents a table c) Describes properties of an entity d) Performs queries	1	CO 1	L1	GATE 2022
11	1	Which normal form ensures no repeating groups? a) Second Normal Form b) Third Normal Form c) First Normal Form d) Boyce-Codd Normal Form	1	CO 1	L6	GATE 2021
12	1	What is the purpose of a "foreign key"? a) To maintain referential integrity between tables b) To identify the primary record c) To delete duplicate data d) To create views	1	CO 1	L1	GATE 2022
13	1	Which database architecture separates user applications from the database?	1	CO 1	L2	GATE 2023

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

		a) One-tier b) Three-tier c) Centralized d) Distributed				
14	1	What is a "schema" in DBMS? a) The data itself b) The structure of the database c) A programming language d) A query	1	CO 1	L3	
15	1	Which operation combines two tables based on a condition? a) Union b) Join c) Select d) Intersect	1	CO 1	L4	GATE 2007
16	1	What does "mapping cardinality" define? a) Number of entities b) Number of entity instances that can be associated c) The keys used in mapping d) Table relationships	1	CO 1	L5	GATE 2012
17	1	Which model uses XML for data representation? a) Relational model b) Network model c) Semi-structured model d) Object-relational model	1	CO 1	L6	
18	1	What ensures transactions are processed reliably? a) DML b) ACID properties c) DDL d) SQL	1	CO 1	L1	
19	1	Which constraint ensures no duplicate values in a column? a) Check b) Foreign Key c) Unique d) Not Null	1	CO 1	L2	
20	1	What is "data redundancy"? a) Storing the same data in multiple places b) Secure backup of data c) Deleting unnecessary data	1	CO 1	L3	GATE 2018

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

		d) Normalizing data				
21	1	Define DBMS and list its main functions.	5	CO 1	L4	
22	1	Explain the three levels of data abstraction in DBMS.	5	CO 1	L5	
23	1	Differentiate between logical and physical data independence.	5	CO 1	L6	
24	1	What are the advantages of DBMS over file systems?	5	CO 1	L1	
25	1	Describe the components of a database system architecture.	5	CO 1	L2	
26	1	Explain the concept of data models with examples.	5	CO 1	L3	
27	1	What is the difference between a schema and an instance?	5	CO 1	L4	
28	1	Describe the entity-relationship model and its components.	5	CO 1	L5	
29	1	What are the different types of database users?	5	CO 1	L6	
30	1	Explain the roles and responsibilities of a DBA.	5	CO 1	L1	
31	1	What is the purpose of database languages?	5	CO 1	L2	
32	1	Differentiate between DDL and DML.	5	CO 1	L3	
33	1	Explain the concept of data redundancy and how DBMS handles it.	5	CO 1	L4	
34	1	What are the different types of database constraints?	5	CO 1	L5	
35	1	Explain the concept of keys in relational databases.	5	CO 1	L6	
36	1	Explain the architecture of a DBMS with a neat diagram and describe each component.	10	CO 1	L1	
37	1	Discuss the advantages and disadvantages of DBMS compared to traditional file systems.	10	CO 1	L2	
38	1	Explain the three-schema architecture of DBMS and its importance in data independence.	10	CO 1	L3	
39	1	Describe the entity-relationship model with examples and explain its components in detail.	10	CO 1	L4	
40	1	Discuss the various types of data models and	10	CO	L5	

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

		compare them with suitable examples.		1		
S.No.	Unit	Question	Marks	CO	BL	Asked in Competitive Exam
UNIT- 2						
1	2	What is the main advantage of using views in databases? a) They store data permanently b) They provide data security and abstraction c) They improve physical data storage d) They reduce disk size	1	CO 2	L5	GATE 2021
2	2	Which component is responsible for concurrency control in DBMS? a) Storage Manager b) Query Processor c) Transaction Manager d) Buffer Manager	1	CO 2	L6	GATE 1995
3	2	What does the term "referential integrity" ensure? a) Valid references between tables are maintained b) Redundant data is removed c) Data is encrypted d) Every table has a primary key	1	CO 2	L1	GATE 2005
4	2	In the three-schema architecture, which schema describes the user view? a) Internal schema b) Conceptual schema c) External schema d) Logical schema	1	CO 2	L2	
5	2	Which of these is a disadvantage of file systems compared to DBMS? a) Ease of access b) High-level abstraction c) Data redundancy and inconsistency d) Data security	1	CO 2	L3	
6	2	What type of relationship exists when one entity is associated with multiple entities of another type? a) One-to-one b) One-to-many c) Many-to-one d) Many-to-many	1	CO 2	L4	
7	2	Which of these is NOT a function of a DBA? a) Managing storage structures	1	CO 2	L5	GATE 2013

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

		b) Backup and recovery c) Writing end-user applications d) Granting access rights				
8	2	What is the purpose of the relational algebra operation "project"? a) Combines rows from two tables b) Filters rows based on condition c) Selects specific columns d) Adds new tuples	1	CO 2	L6	
9	2	In the ER model, what describes the characteristics of an entity? a) Entity set b) Relationship c) Attributes d) Schema	1	CO 2	L1	
10	2	Which of these is a procedural query language? a) SQL b) Relational Algebra c) QBE d) Datalog	1	CO 2	L2	
11	2	What is the main purpose of normalization? a) To eliminate redundancy and improve integrity b) To reduce number of queries c) To enforce access control d) To increase complexity	1	CO 2	L3	
12	2	Which type of data model is most commonly used in modern DBMS? a) Network model b) Hierarchical model c) Relational model d) Object-oriented model	1	CO 2	L4	GATE 2017
13	2	What does the term "atomicity" refer to in transactions? a) All users see consistent data b) A transaction is either fully completed or not at all c) Data remains permanent after commit d) Concurrent transactions do not conflict	1	CO 2	L5	GATE 2006
14	2	Which of these is NOT a database constraint? a) Primary key b) Foreign key	1	CO 2	L6	

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

		c) Index d) Not null				
15	2	What is the purpose of a data dictionary? a) To store metadata about database structure b) To back up the database c) To log user transactions d) To manage concurrent access	1	CO 2	L1	
16	2	Which of these is a physical data independence example? a) Changing table schema b) Adding new views c) Changing file organization of stored data d) Modifying application logic	1	CO 2	L2	GATE 2021
17	2	What is the main purpose of a surrogate key? a) To enforce referential integrity b) To uniquely identify a record when natural key is not available c) To establish relationships d) To reduce table size	1	CO 2	L3	GATE 2022
18	2	Which of these is NOT a component of a DBMS? a) Query processor b) Compiler c) Transaction manager d) Storage manager	1	CO 2	L4	GATE 2023
19	2	What is the main advantage of using views in databases? a) They store data permanently b) They provide data security and abstraction c) They improve physical data storage d) They reduce disk size	1	CO 2	L5	GATE 2024
20	2	Which component is responsible for concurrency control in DBMS? a) Storage Manager b) Query Processor c) Transaction Manager d) Buffer Manager	1	CO 2	L6	GATE 2020
21	2	What is the purpose of normalization in database design?	5	CO 2	L1	
22	2	Describe the three-schema architecture of DBMS.	5	CO 2	L2	
23	2	What are the ACID properties of transactions?	5	CO	L3	

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

				2		
24	2	Explain the concept of mapping cardinalities in ER diagrams.	5	CO 2	L4	
25	2	What is the difference between strong and weak entities?	5	CO 2	L5	
26	2	Explain the concept of derived attributes with examples.	5	CO 2	L6	
27	2	What are the different types of relationships in ER modeling?	5	CO 2	L1	
28	2	Explain the purpose of views in databases.	5	CO 2	L2	
29	2	What is the difference between procedural and non-procedural DML?	5	CO 2	L3	
30	2	Explain the concept of data dictionary.	5	CO 2	L4	
31	2	What are the different types of data models?	5	CO 2	L5	
32	2	Explain the concept of generalization and specialization.	5	CO 2	L6	
33	2	What is the difference between logical and physical database design?	5	CO 2	L1	
34	2	Explain the concept of referential integrity.	5	CO 2	L2	
35	2	What are the different types of database end users?	5	CO 2	L3	
36	2	Discuss the role of reinforcement learning in real-world applications.	10	CO 2	L4	
37	2	Describe the time and space complexity of feature scaling in preprocessing.	10	CO 2	L5	
38	2	Explain the implementation of stratified k-fold cross-validation with an example.	10	CO 2	L6	
39	2	Discuss the role of data normalization in improving model performance.	10	CO 2	L1	
40	2	Describe the applications of feature engineering in real-world scenarios.	10	CO 2	L2	
S.No.	Unit	Question	Mark s	CO	B L	Asked in Competitive Exam
UNIT-3						
1	3	What does the term "null" represent in databases? a) A value of zero b) A missing or unknown value	1	CO 3	L2	

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

		c) A duplicate value d) A default value				
2	3	What does the term "atomicity" refer to in transactions? a) Splitting transactions into parts b) Running multiple transactions at once c) Ensuring transactions are all-or-nothing d) Combining transactions together	1	CO 3	L3	
3	3	Which operation selects tuples that satisfy a given predicate? a) Union b) Select (σ) c) Project (π) d) Join	1	CO 3	L4	
4	3	The ____ operation selects certain columns from a table and discards others. a) Join b) Project (π) c) Select (σ) d) Union	1	CO 3	L5	GATE 2000
5	3	Which operation combines two relations and returns all tuples that are in either relation? a) Intersection b) Cartesian Product c) Union d) Difference	1	CO 3	L6	GATE 2008
6	3	The set-difference operation returns tuples that are in: a) Both relations b) The second relation but not the first c) The first relation but not the second d) Neither relation	1	CO 3	L1	
7	3	The Cartesian product of two relations R and S is denoted by: a) $R \cup S$ b) $R \cap S$ c) $R \times S$ d) $R - S$	1	CO 3	L2	
8	3	The ____ operation is used to rename relations or attributes. a) Union	1	CO 3	L3	

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

		b) Join c) Intersection d) Rename (ρ)				
9	3	Which operation combines related tuples from two relations into single tuples? a) Select b) Project c) Join d) Rename	1	CO 3	L4	GATE 2021
10	3	The natural join operation performs a(n): a) Arbitrary combination of tuples b) Cartesian product c) Join based on common attributes d) Aggregation of results	1	CO 3	L5	
11	3	Which operation assigns a temporary name to a relation? a) Join b) Select c) Rename (ρ) d) Union	1	CO 3	L6	
12	3	The outer join operation preserves: a) Only matched tuples b) Only unmatched tuples c) All matched and unmatched tuples d) Only primary keys	1	CO 3	L1	
13	3	Which of the following is a fundamental operation in relational algebra? a) Sort b) Select (σ) c) Aggregate d) Index	1	CO 3	L2	
14	3	The ____ operation removes duplicate tuples from a relation. a) Project (π) b) Select (σ) c) Join d) Union	1	CO 3	L3	
15	3	Which operation returns tuples that are in both relations? a) Union b) Difference	1	CO 3	L4	GATE 1997

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

		c) Intersection d) Cartesian Product				
16	3	The result of a relational algebra operation is always a) A number b) A set of functions c) A relation d) A string	1	CO 3	L5	GATE 2021
17	3	Which operation is used to retrieve data from multiple relations? a) Rename b) Join c) Project d) Select	1	CO 3	L6	
18	3	The ____ operation divides tuples of one relation by another. a) Union b) Intersection c) Division ($\div$) d) Selection	1	CO 3	L1	
19	3	Which operation is used to compute aggregate functions like SUM, AVG? a) Project b) Join c) Select d) Aggregate (γ)	1	CO 3	L2	
20	3	What does the term "null" represent in databases? a) A value of zero b) A missing or unknown value c) A duplicate value d) A default value	1	CO 3	L3	GATE 2020
21	3	Explain the concept of database system utilities.	5	CO 3	L4	
22	3	What are the different types of database interfaces?	5	CO 3	L5	
23	3	Explain the concept of data storage and indexing.	5	CO 3	L6	
24	3	What are the different types of database system architectures?	5	CO 3	L1	
25	3	Explain the concept of query processing.	5	CO 3	L2	
26	3	What are the different types of database languages?	5	CO 3	L3	

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

27	3	Explain the concept of transaction management.	5	CO 3	L4	
28	3	What are the different types of database constraints?	5	CO 3	L5	
29	3	Explain the concept of database security.	5	CO 3	L6	
30	3	What are the different types of database recovery techniques?	5	CO 3	L1	
31	3	Define the Select operation in relational algebra.	5	CO 3	L2	
32	3	Explain the Project operation with an example.	5	CO 3	L3	
33	3	Differentiate between Union and Set-Difference operations.	5	CO 3	L4	
34	3	What is the purpose of the Cartesian Product operation?	5	CO 3	L5	
35	3	Describe the Rename operation and its significance.	5	CO 3	L6	
36	3	Explain the concept of normalization and its importance in database design.	10	CO 3	L1	
37	3	Describe the ACID properties of transactions and their importance in database systems.	10	CO 3	L2	
38	3	Discuss the various types of database languages and their functions in DBMS.	10	CO 3	L3	
39	3	Explain the concept of data independence and its types with suitable examples.	10	CO 3	L4	
40	3	Describe the various types of database system architectures with their advantages and disadvantages.	10	CO 3	L5	
S.No.	Unit	Question	Marks	CO	B L	Asked in Competitive Exam
UNIT-4						
1	4	The ____ operation extends the relational algebra with arithmetic operations. a) Selection b) Extended Projection c) Natural Join d) Intersection	1	CO 4	L5	GATE 2002
2	4	Which operation returns all possible combinations of tuples from two relations?	1	CO 4	L6	GATE 2014

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

		a) Natural Join b) Inner Join c) Cartesian Product d) Selection				
3	4	The ____ operation is used to store intermediate results. a) Join b) Projection c) Assignment d) Division	1	CO 4	L1	
4	4	Which operation removes columns from a relation? a) Selection b) Join c) Projection d) Union	1	CO 4	L2	
5	4	The ____ operation returns only distinct tuples from a relation. a) Selection b) Assignment c) Projection d) Cartesian Product	1	CO 4	L3	GATE 2020
6	4	Which operation is used to rename attributes in a relation? a) Join b) Rename c) Union d) Intersection	1	CO 4	L4	
7	4	The ____ operation combines two relations and keeps only matching tuples. a) Cartesian Product b) Natural Join c) Union d) Outer Join	1	CO 4	L5	
8	4	Which operation is used to filter rows based on a condition? a) Selection b) Projection c) Join d) Union	1	CO 4	L6	
9	4	The ____ operation returns tuples from one relation that do not appear in another.	1	CO 4	L1	

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

		a) Join b) Intersection c) Set Difference d) Projection				
10	4	Which operation is used to compute statistical summaries? a) Projection b) Aggregation c) Selection d) Rename	1	CO 4	L2	GATE 2020
11	4	The ____ operation allows arithmetic expressions in the projection list. a) Join b) Selection c) Extended Projection d) Union	1	CO 4	L3	GATE 2010
12	4	Which operation is used to combine relations without losing unmatched tuples? a) Outer Join b) Natural Join c) Intersection d) Cartesian Product	1	CO 4	L4	GATE 2019
13	4	The ____ operation is used to define a temporary relation. a) Selection b) Join c) Assignment d) Projection	1	CO 4	L5	GATE 2013
14	4	Which operation returns only the specified columns from a relation? a) Projection b) Selection c) Union d) Join	1	CO 4	L6	
15	4	The ____ operation removes duplicates from the result of a projection. a) Cartesian Product b) Selection c) Join d) Projection	1	CO 4	L1	
16	4	Which operation is used to rename a relation or its	1	CO	L2	

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

		attributes? a) Join b) Rename c) Assignment d) Union		4		
17	4	The ____ operation combines two relations and includes NULL for non-matching tuples. a) Intersection b) Natural Join c) Outer Join d) Cartesian Product	1	CO 4	L3	GATE 2022
18	4	Which operation is used to filter tuples based on a condition? a) Selection b) Projection c) Join d) Rename	1	CO 4	L4	
19	4	The ____ operation returns tuples present in one relation but not in another. a) Intersection b) Set Difference c) Join d) Projection	1	CO 4	L5	
20	4	Which operation is used to compute SUM, AVG, COUNT, etc.? a) Projection b) Join c) Rename d) Aggregation	1	CO 4	L6	
21	4	Explain the Natural Join operation with an example.	5	CO 4	L1	
22	4	What is the role of the Assignment operation in relational algebra?	5	CO 4	L2	
23	4	How does Outer Join differ from Natural Join?	5	CO 4	L3	
24	4	List the fundamental operations of relational algebra.	5	CO 4	L4	
25	4	Explain Generalized Projection with an example.	5	CO 4	L5	
26	4	What is the purpose of Aggregation in relational algebra?	5	CO 4	L6	

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

27	4	Differentiate between Theta Join and Equijoin.	5	CO 4	L1	
28	4	Describe the Division operation with an example.	5	CO 4	L2	
29	4	What are the extended operations in relational algebra?	5	CO 4	L3	
30	4	Explain Set-Intersection with an example.	5	CO 4	L4	
31	4	How does Relational Algebra support query optimization?	5	CO 4	L5	
32	4	What is the difference between Select and Project operations?	5	CO 4	L6	
33	4	Describe the Full Outer Join operation.	5	CO 4	L1	
34	4	Why is duplicate elimination important in Project operations?	5	CO 4	L2	
35	4	Explain the syntax of a relational algebra expression.	5	CO 4	L3	
36	4	Explain all fundamental operations of relational algebra with examples.	10	CO 4	L4	
37	4	Compare and contrast Natural Join and Theta Join with examples.	10	CO 4	L5	
38	4	Describe the extended relational algebra operations and their significance.	10	CO 4	L6	
39	4	How does aggregation enhance relational algebra? Explain with examples.	10	CO 4	L1	
40	4	Discuss the Outer Join operation and its variants with examples.	10	CO 4	L2	
S.No.	Unit	Question	Marks	CO	B L	Asked in Competitive Exam
UNIT-5						
1	5	The ____ operation stores intermediate query results. a) Project b) Join c) Temporary Table d) Group	1	CO 5	L2	GATE 2021
2	5	Which of the following best describes a relational database? a) A spreadsheet with unrelated data b) A collection of related tables	1	CO 5	L3	GATE 2012

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

		c) A hierarchical data structure d) A file system				
3	5	What is a relation in the relational model? a) A single record b) A table with rows and columns c) A column in a table d) A pointer to another table	1	CO 5	L4	GATE 2011
4	5	What is a tuple in a relational database? a) A table b) A row in a table c) A column d) A key	1	CO 5	L5	
5	5	What is an attribute in a relational database? a) A row b) A column in a table c) A relationship d) A schema	1	CO 5	L6	
6	5	Which term refers to the number of attributes in a relation? a) Cardinality b) Degree c) Arity d) Dimension	1	CO 5	L1	
7	5	What is a domain in the relational model? a) A set of valid values for an attribute b) A type of relationship c) A tuple identifier d) A group of tables	1	CO 5	L2	GATE 2022
8	5	What is a database schema? a) Data stored in a table b) The structure that defines tables, attributes, and relationships c) A query output d) A user interface	1	CO 5	L3	GATE 2012
9	5	Which of the following is a superkey? a) A set of attributes that uniquely identify a tuple b) A column with duplicate values c) A foreign key d) A null value column	1	CO 5	L4	GATE 2004
10	5	What is a candidate key? a) A key that always has NULL values	1	CO 5	L5	

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

		b) A foreign key c) A minimal superkey d) A combination of all columns				
11	5	What is a primary key? a) A non-unique identifier b) A duplicated column c) A unique identifier for tuples in a table d) A table name	1	CO 5	L6	
12	5	What is a foreign key? a) A key defined in the same table b) A key that refers to the primary key of another table c) A duplicate attribute d) A NULL value	1	CO 5	L1	
13	5	Which constraint ensures that a column cannot have NULL values? a) UNIQUE b) NOT NULL c) DEFAULT d) CHECK	1	CO 5	L2	GATE 2023
14	5	What is referential integrity? a) Ensuring primary key uniqueness b) Ensuring foreign key values match primary key values in another table c) Preventing duplicate rows d) Normalizing data	1	CO 5	L3	
15	5	What is a schema diagram? a) A visual representation of database structure b) A result of a query c) A table content d) A SQL script	1	CO 5	L4	
16	5	Which of the following is NOT a key in a relational database? a) Primary key b) Candidate key c) Foreign key d) Secondary key	1	CO 5	L5	
17	5	What is the difference between a superkey and a candidate key? a) A candidate key is larger than a superkey b) A superkey must have null values	1	CO 5	L6	

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

		c) A candidate key is a minimal superkey d) No difference at all				
18	5	What is a composite key? a) A single-column key b) A key made up of two or more attributes c) A foreign key d) A default constraint	1	CO 5	L1	GATE 1999
19	5	Which of the following ensures entity integrity? a) Foreign key b) Primary key c) Composite key d) NULL constraint	1	CO 5	L2	
20	5	What is the purpose of a foreign key constraint? a) To define primary keys b) To store binary data c) To maintain referential integrity between tables d) To allow NULL values	1	CO 5	L3	GATE 2020
21	5	What is a lossless join decomposition?	5	CO 5	L4	
22	5	How does the Semi-Join operation work?	5	CO 5	L5	
23	5	Explain the difference between Left Outer Join and Right Outer Join.	5	CO 5	L6	
24	5	What is the role of the Rename operation in complex queries?	5	CO 5	L1	
25	5	How does Relational Algebra handle NULL values in operations?	5	CO 5	L2	
26	5	Describe the use of the Division operation in real-world queries.	5	CO 5	L3	
27	5	What is the difference between a Derived Relation and a Base Relation?	5	CO 5	L4	
28	5	Explain how Aggregation functions (SUM, AVG) work in relational algebra.	5	CO 5	L5	
29	5	What is the significance of the Closure Property in relational algebra?	5	CO 5	L6	
30	5	How does the Assignment operation aid in query readability?	5	CO 5	L1	
31	5	Explain the difference between a Natural Join and an Equijoin.	5	CO 5	L2	
32	5	What are the limitations of the Cartesian Product operation?	5	CO 5	L3	

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

33	5	Describe how Outer Joins preserve unmatched tuples.	5	CO 5	L4	
34	5	Why is the Rename operation essential for self-joins?	5	CO 5	L5	
35	5	Explain the concept of "relation compatibility" in Union operations.	5	CO 5	L6	
36	5	Discuss the Assignment operation and its use in complex queries.	10	CO 5	L1	
37	5	How does relational algebra handle duplicate elimination?	10	CO 5	L2	
38	5	Explain the Set-Intersection operation with an example.	10	CO 5	L3	
39	5	What are the limitations of relational algebra compared to SQL?	10	CO 5	L4	
40	5	How does nested relational algebra work?	10	CO 5	L5	

Signature of Faculty/ Course Coordinator
Dean / Head of Department